The margin–sensitivity coupling: what is proved, why the impossibility reading was wrong, and three conjectures that would close it

Working note (isolated from the main paper for independent attack)

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Abstract

This note isolates the "self-limiting coupling" theory that appears as Theorem 3 (`thm:coupling`) in the main paper, so it can be examined and, ideally, strengthened in isolation. It records (i) the one inequality that is genuinely *proved* (a homogeneity lower envelope on the margin gradient); (ii) a precise account of *why* the original "η/L cannot be trained" *impossibility* reading does not follow from that inequality; and (iii) three conjectures, each of which, if proved, would upgrade the loose envelope into a real obstruction to training the certified ratio. The proved part is short and correct; the open part is where the contribution would be. Nothing here is claimed as a theorem beyond Theorem 1 and Lemma 1; everything labelled Conjecture is open.

1 Setup and the quantity at stake

Let $f : \mathbb{R}^d \to \mathbb{R}^k$ be a classifier with logits $f_1, \ldots, f_k$, and for a labelled point (x, y) define the *signed logit margin*

$$M(x) \;=\; f_y(x) - \max_{j \neq y} f_j(x).$$

$M(x) > 0$ iff f is correct at x. Write $\eta = \mathbb{E}[M(x)]$ for the mean margin over correctly classified data and $L = \mathbb{E}\,\|\nabla M(x)\|$ for the mean input-gradient norm (the local sensitivity), with the norm chosen dual to the threat model (ℓ_2 certificate uses $\|\cdot\|_2$; an ℓ_∞ threat uses $\|\cdot\|_1$). The first-order certified radius is

$$r_2(f, x) \;\geq\; \frac{M(x)}{\|\nabla M(x)\|_2} \qquad \text{(Hein–Andriushchenko 2017; Tsuzuku et al. 2018)}, \tag{1}$$

so the dataset ratio η/L is a natural robustness surrogate. The empirical finding of the paper is that η/L *diagnoses* robustness very well across architectures, yet *training* it (penalizing $\|\nabla M\|$, or maximizing the ratio) does not buy robustness. The question this note isolates is: *is there a theorem behind that failure, or only an empirical regularity?*

2 What is proved: a homogeneity lower envelope

Theorem 1 (Homogeneous self-limiting envelope)**.** *Let f be* bias-free *and let every activation be positively homogeneous of degree one ($\sigma(ct) = c\,\sigma(t)$ for $c > 0$; ReLU, leaky-ReLU, absolute value, maxout). Then M is positively homogeneous of degree one, and at every differentiability point x,*

$$\langle \nabla M(x),\, x \rangle \;=\; M(x) \quad \textit{(Euler's identity)}, \tag{2}$$

hence, by Cauchy–Schwarz and Hölder,

$$\|\nabla M(x)\|_2 \;\geq\; \frac{M(x)}{\|x\|_2}, \qquad \|\nabla M(x)\|_1 \;\geq\; \frac{M(x)}{\|x\|_\infty}. \tag{3}$$

Consequently the per-sample ratio is capped by the input norm, $M(x)/\|\nabla M(x)\|_2 \leq \|x\|_2$, and the dataset ratio obeys $\eta/L \leq \sup_i \|x_i\|_2$, independently of training.

Proof. Each logit is an alternating composition of bias-free linear maps and degree-1 positively homogeneous activations, so $f_j(cx) = c\, f_j(x)$ for $c > 0$ by induction on depth; a pointwise maximum of degree-1 positively homogeneous functions is again degree-1 positively homogeneous, so $M(cx) = cM(x)$. Differentiating in c at $c = 1$ gives (2) wherever M is differentiable (a.e. by Rademacher; on the measure-zero nonsmooth set of a ReLU network a Clarke subgradient v with $\langle v, x\rangle = M(x)$ exists). Then $M(x) = \langle \nabla M(x), x\rangle \leq \|\nabla M(x)\|_2 \|x\|_2$ and the Hölder version give (3); rearranging gives the cap. □

Remark 1. Theorem 1 is correct as stated. It is essentially Euler's identity for 1-homogeneous functions plus Cauchy–Schwarz; the content is only the *interpretation* that the margin gradient has an architecture-imposed floor $M(x)/\|x\|_2$ that scales with the margin. This much *is* a structural statement: the coupling between margin and sensitivity is not purely an optimization artifact, because no bias-free homogeneous network can have a vanishing gradient at a point of nonzero margin.

3 Why the impossibility reading was wrong

The main paper originally read Theorem 1 as an *impossibility*: "η/L cannot be trained, so no η/L-guided defense can exist." That reading does not follow from (3). Four distinct gaps, in increasing order of importance:

(W1) The bound points the wrong way to forbid training. To *raise* the certified ratio $\eta/L = M/\|\nabla M\|$ one wants the denominator $\|\nabla M\|$ *small* (or the margin large). Inequality (3) is a *lower* bound on $\|\nabla M\|$, equivalently an *upper* cap $\eta/L \leq \|x\|$. An upper cap does not forbid increasing a quantity that currently sits *below* the cap; it only says how far the increase could go. A genuine no-go needs a statement that η/L is already *at* (a constant fraction of) its ceiling, or that increasing it does not help robustness. Neither is in (3).

(W2) The envelope is loose at the operating point. Empirically, for the trained networks the realized gradient norm sits far above the floor: $\|\nabla M(x)\|_2 \approx 26 \times M(x)/\|x\|_2$ at the data (the floor is a few percent of the measured value). So (3) permits the gradient to be reduced by more than an order of magnitude — i.e. η/L to be raised by more than an order of magnitude — before the bound binds. A bound with that much slack cannot, by itself, explain why a penalty fails to raise robustness.

(W3) The hypotheses fail for the real networks. Identity (2) needs *bias-free* and *exactly* degree-1 homogeneous architectures. The experimental models are PreActResNet-18 with additive biases and batch normalization (and, in some controls, smooth activations), all of which break exact 1-homogeneity. On those models Theorem 1 is at best an idealized mechanism, not an exact law, so it cannot carry an impossibility claim about *them.*

(W4) Static identity vs. dynamic lockstep. The empirical obstruction is dynamical: along a regularization path $\min \mathcal{L}_{\mathrm{AT}} + \lambda \mathbb{E}\|\nabla M\|$, the margin η falls in *lockstep* with L, so the ratio barely moves and the certified radius does not improve. Theorem 1 is a *static* inequality at a fixed network; it says nothing about how (η, L) co-move under a training penalty. The original error was to treat a static inequality as if it constrained the training *trajectory*.

What survives rigorously after the walk-back is exactly two facts:

Lemma 1 (Scale degeneracy of the ratio)**.** *The per-sample ratio $M(x)/\|\nabla M(x)\|$ is invariant under $f \mapsto cf$ for $c > 0$. Hence the objective "maximize η/L" (equivalently "penalize $\|\nabla M\|/M$ on correct points") is blind to the absolute margin and admits the constant classifier $\nabla M \equiv 0$ as a global optimizer. Direct ratio maximization is therefore ill-posed.*

Proof. Under $f \mapsto cf$ both M and ∇M scale by c, leaving the ratio fixed; the penalty $\mathbb{E}[\|\nabla M\|/M] \geq 0$ attains its infimum 0 only as $\nabla M \to 0$, i.e. a constant score (chance accuracy). This is the constant- function degeneracy noted by Tsuzuku et al. (2018). □

Lemma 1 kills the *ratio-maximization* route cleanly, but it says nothing about the *margin-gradient penalty* route (W4); that one is, at present, only an empirical regularity (the four interventions). The honest current status is: ratio maximization is provably degenerate (Lemma 1); the gradient-penalty lockstep is observed, not proved; and the homogeneity envelope (Theorem 1) is a true but loose structural floor that does *not* forbid training.

4 Three conjectures that would close the gap

Each conjecture below, if proved, converts a piece of the empirical story into a theorem. They are ordered by how directly they would restore an impossibility claim. Throughout, "the trained solution" means the directional limit of gradient flow on the logistic/exponential loss for a homogeneous network, which converges to a KKT point of the max-margin program (Lyu–Li 2020; Ji–Telgarsky).

Conjecture A: a matching upper envelope at the implicit-bias solution

Theorem 1 gives a lower envelope $\|\nabla M(x)\|_2 \geq M(x)/\|x\|_2$. The missing half is an *upper* envelope *at the GD-selected solution.*

Conjecture 1 (Tight coupling at max margin)**.** Let f be a bias-free, 1-homogeneous network and let $f^\star$ be the max-margin direction reached by gradient flow on a separable dataset $\{(x_i, y_i)\}$. Then there is a constant $\kappa \geq 1$, depending only on the data geometry (e.g. the margin condition number / the number of active support neurons), not on the width, such that at every support point x_i

$$\frac{M(x_i)}{\|x_i\|_2} \;\leq\; \|\nabla M(x_i)\|_2 \;\leq\; \kappa\,\frac{M(x_i)}{\|x_i\|_2}.$$

Consequently $\eta/L \in [\|x\|/\kappa,\ \|x\|]$ is pinned to a $\Theta(\|x\|)$ band at the trained solution, and no reparametrization or first-order penalty that stays at a max-margin KKT point can move it outside that band.

Why this would close it. An upper envelope of the same order as the lower one (W2 says the realized $\kappa \approx 26$; the conjecture asks whether κ is *bounded* by data geometry rather than free) means η/L is essentially *determined* by $\|x\|$ at any max-margin solution, so the only way to raise it is to change the data scaling or the capacity — exactly the impossibility the paper wanted.

Suggested attack. At a max-margin KKT point, $\nabla M(x_i) = \sum_{r\in\mathcal{A}(x_i)} a_r w_r$ over active neurons; bound $\|\nabla M(x_i)\|$ above using the KKT stationarity $\sum_i \lambda_i y_i \nabla_\theta f(x_i) = \theta$ and the homogeneity relation $\|\theta\|^2 = \sum_i \lambda_i y_i M(x_i)$ to control the number/coherence of active neurons. The danger case is highly anisotropic data where κ blows up (the $t \mapsto t^{2n+1}$ example in §5 has $\kappa \to \infty$), so the conjecture must restrict to a margin/condition assumption.

Conjecture B: penalty-path lockstep (no first-order decoupling)

This formalizes (W4) and the four-intervention experiment directly.

Conjecture 2 (Margin falls with the gradient along the penalty path)**.** Consider the regularized adversarial objective $\mathcal{L}(\theta) + \lambda\, R(\theta)$ with $R(\theta) = \mathbb{E}\|\nabla_x M\|_q$ and let θ_λ be its minimizer (or a gradient-flow path in λ). Write $\eta(\lambda) = \mathbb{E}[M]$ and $L(\lambda) = \mathbb{E}\|\nabla_x M\|_q$ at θ_λ. Then there is a constant $c > 0$ with

$$\frac{d\eta}{d(\log L)} \;\geq\; c\,\eta \qquad \text{(equivalently } \tfrac{d}{d\lambda}\log\eta \;\geq\; c\,\tfrac{d}{d\lambda}\log L\text{)},$$

so that η/L^c is non-decreasing while η/L is (to first order) stationary: reducing the sensitivity by a factor reduces the margin by at least a comparable factor, and the certified ratio does not improve.

Why this would close it. It turns "every penalty we tried collapsed the margin" into a statement about *all* first-order penalties of this form. *Suggested attack.* Differentiate the KKT/stationarity conditions of the penalized program in λ (an implicit-function / envelope-theorem calculation): the first-order change in η is governed by the same Hessian block as the change in L because both are functionals of $\nabla_x M$; show the cross term forces co-movement. A clean toy case to prove first: a single homogeneous neuron / a linear-in-features model where the penalty and the margin share a gradient direction.

Conjecture C: a local, noise-free capacity ceiling

The capacity remark in the paper invokes Bubeck–Sellke (2021): any p-parameter interpolator of n points in $\mathbb{R}^d$ on isoperimetric data *with label noise* has *global* Lipschitz constant $\tilde{\Omega}(\sqrt{nd/p})$, which gives $\eta/L \leq \tilde{O}(\sqrt{p/(nd)})$ *for the global constant.* Two caveats weaken it: it bounds the global Lipschitz constant, not the local gradient norm $L = \mathbb{E}\|\nabla M\|$ that we measure; and it assumes label noise.

Conjecture 3 (Local capacity bound)**.** Under the data assumptions of Bubeck–Sellke but *without* requiring label noise, and with L the *local* expected margin-gradient norm at the data rather than the global Lipschitz constant, there is still a bound of the form $\eta/L \leq \tilde{O}\big(\sqrt{p/(nd)}\big)$ up to logarithmic factors.

Why this would close it. It would make the capacity ceiling a theorem about the *measured* ratio, explaining the architecture-independence of the coupling (ReLU/SiLU/GELU alike) from first principles rather than by analogy. *Suggested attack.* Replace the global-Lipschitz step in Bubeck–Sellke by a local Poincaré/concentration argument at the data measure; the label-noise assumption may be replaceable by a margin/anti-concentration assumption on M.

(Optional) Conjecture D: the implicit bias controls the bracket condition number

The two-sided bracket (§5) shows $\eta/L \leq r_2 \leq \eta/\alpha$ with $\kappa = L/\alpha \geq 1$, exact ($\kappa = 1$) for invariant linear features. A bound on κ at the trained solution would tie the diagnostic to the true radius.

Conjecture 4. For the max-margin solution of a homogeneous network on data with margin condition μ, the bracket condition number satisfies $\kappa = L/\alpha \leq g(\mu)$ for an explicit g, so η/L estimates r_2 up to the factor $g(\mu)$.

5 Reference results used above (for self-containment)

The two-sided bracket. For a score g correct on x with margin $\eta = y\,g(x) > 0$, if g is L-Lipschitz and has co-Lipschitz (lower-Lipschitz) constant α along a path to the boundary on a convex neighbourhood, then $\eta/L \leq r_2 \leq \eta/\alpha$, with $\kappa := L/\alpha \geq 1$ and equality $r_2 = \eta/L$ iff $\kappa = 1$. For a linear invariant feature $g(x) = w^\top x + b$ with w in the invariant subspace, $L = \alpha = \|w\|_2$, so $\kappa = 1$ and $r_2 = \eta/\|w\|_2$ exactly. The unbounded-κ end is the feature $t \mapsto t^{2n+1}$ on $\mathbb{R}$: $\eta/L = 1/(2n+1)$ while $r_2 = 1$, so no *universal* converse exists.

Implicit bias to max margin. For homogeneous networks, gradient flow on the logistic/exponential loss converges in direction to a KKT point of $\min \|\theta\|$ s.t. $y_i f_\theta(x_i) \geq 1$ (Lyu–Li 2020; Ji–Telgarsky 2020). This is what makes "the trained solution" a well-defined object for Conjectures 1–2.

Status summary

Statement	Status	Where
Lower envelope $\|\nabla M\|_2 \geq M/\|x\|_2$	**proved**	Thm 1
Ratio maximization is degenerate	**proved**	Lem 1
"η/L cannot be trained" (impossibility)	**withdrawn**	§3
Matching upper envelope at max margin	*conjecture*	Conj 1
Penalty-path margin/gradient lockstep	*conjecture*	Conj 2
Local, noise-free capacity ceiling	*conjecture*	Conj 3
Bracket condition number bound	*conjecture*	Conj 4

Proving any one of A, B, or C would restore a genuine (not loose, not merely empirical) obstruction to first-order training of η/L, and would be the theoretical contribution the walked-back Theorem 3 only gestured at.